Ethanol Blending in Gasoline – Lithuania

September 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Europe

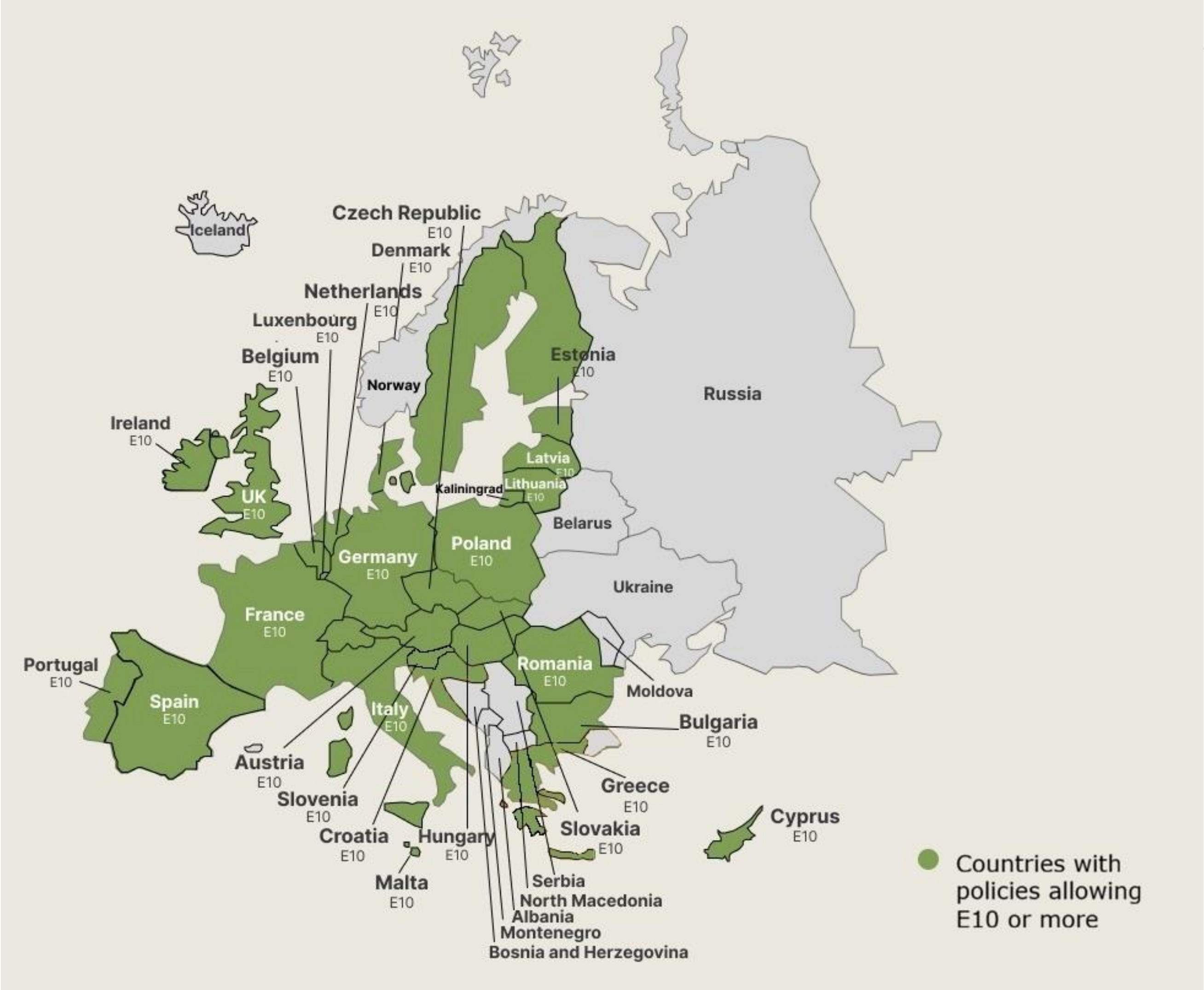
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.

- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.

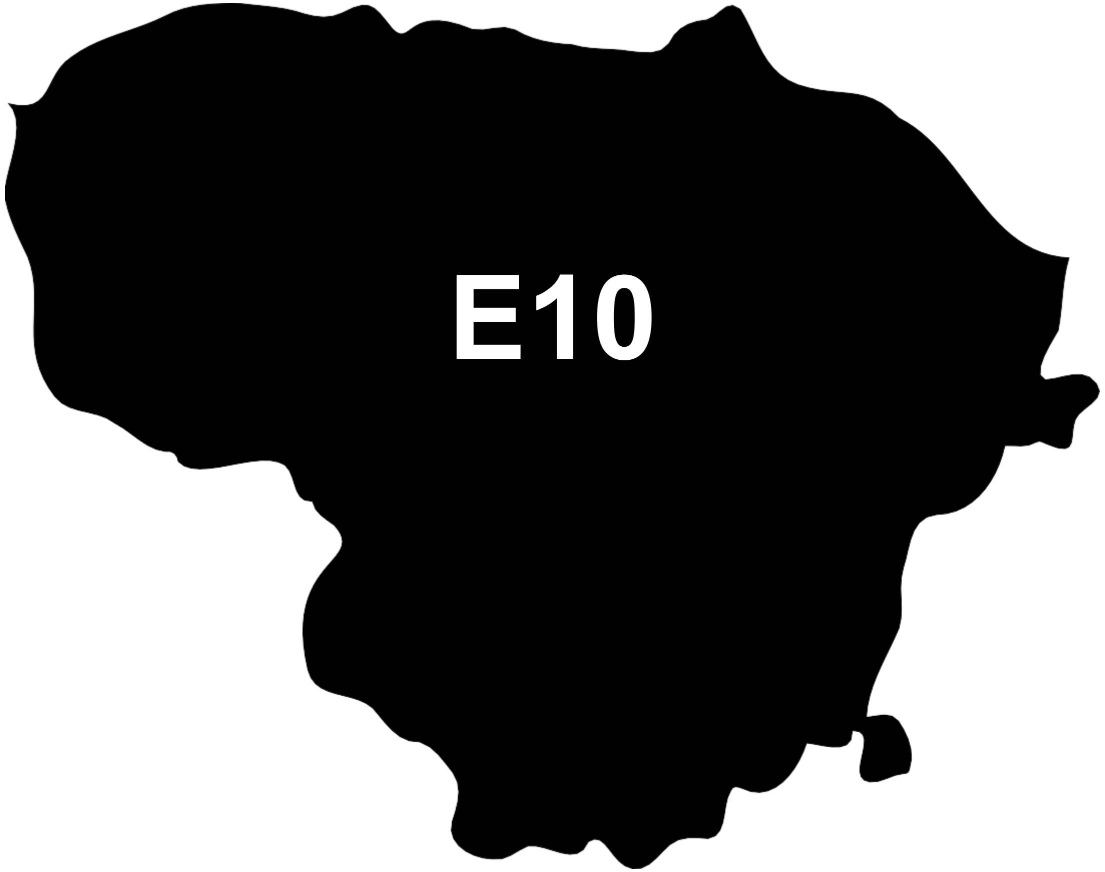
- Ethanol contributes to transport decarbonization and air quality improvement.

- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Lithuania

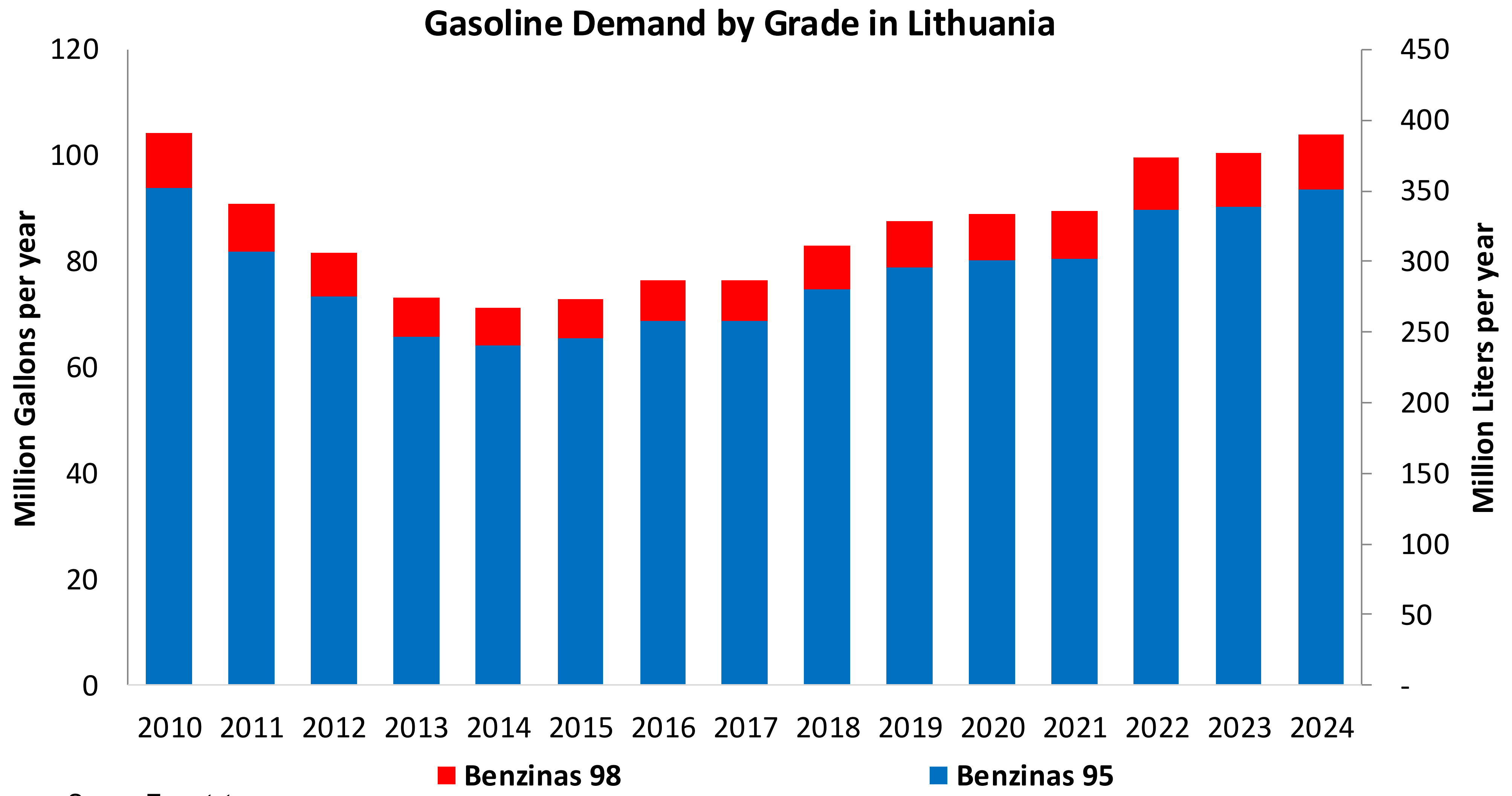


In 2024, gasoline consumption in Lithuania was 390 million liters (103 million gallons) and growth rate was 3.5%. Gasoline production and net imports were 3,762 and -3,361 million liters (994 and -888 million gallons) correspondingly. Lithuania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

In 2024, ethanol consumption in Lithuania was 45 million liters (12 million gallons) representing 11.4% of gasoline demand. Ethanol production and Net Imports were 26 and 19 million liters (12 and 5 million gallons) correspondingly. Ethanol sources are Corn 15% and Cereals 85%.

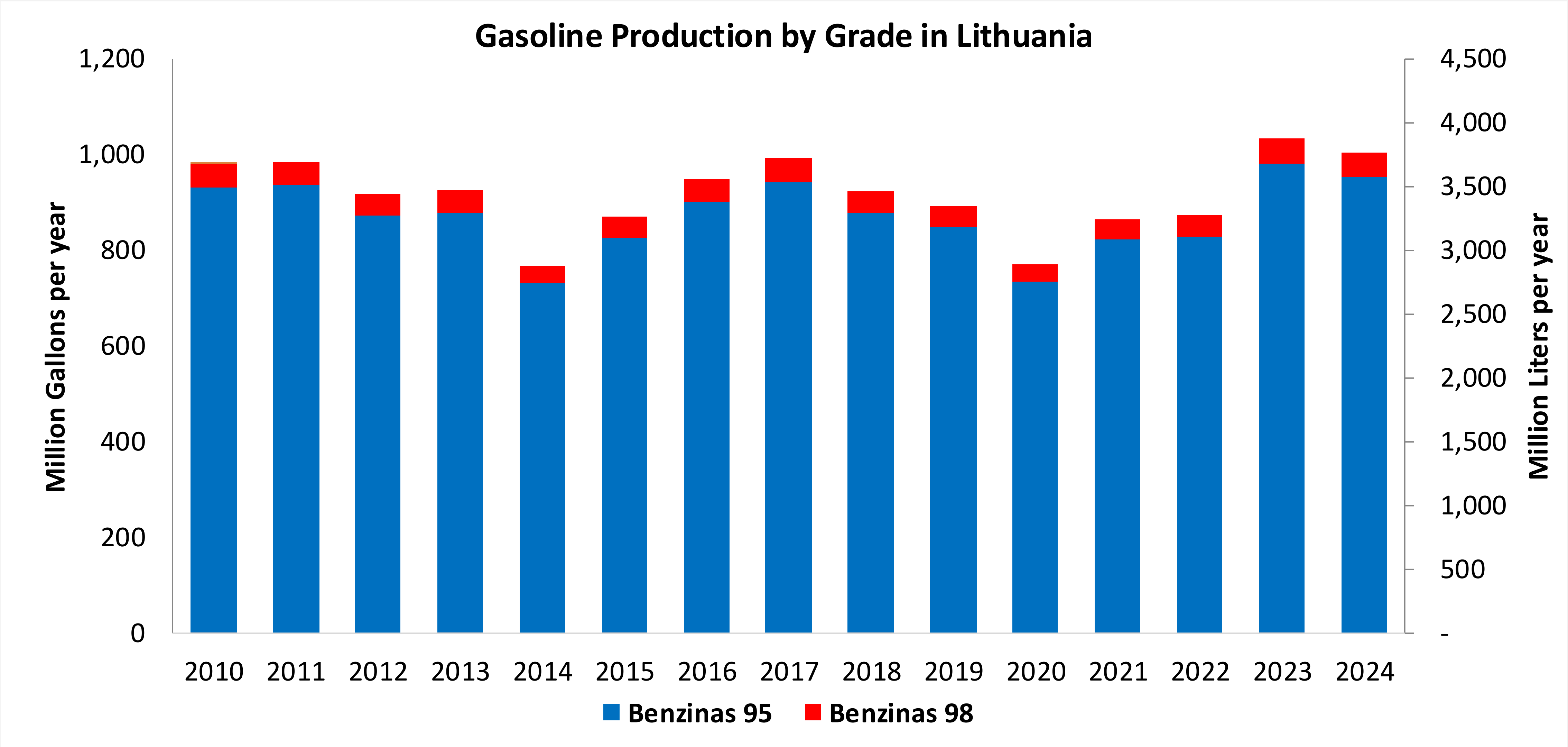
Source: Eurostat, European Commision

Gasoline Demand in Lithuania



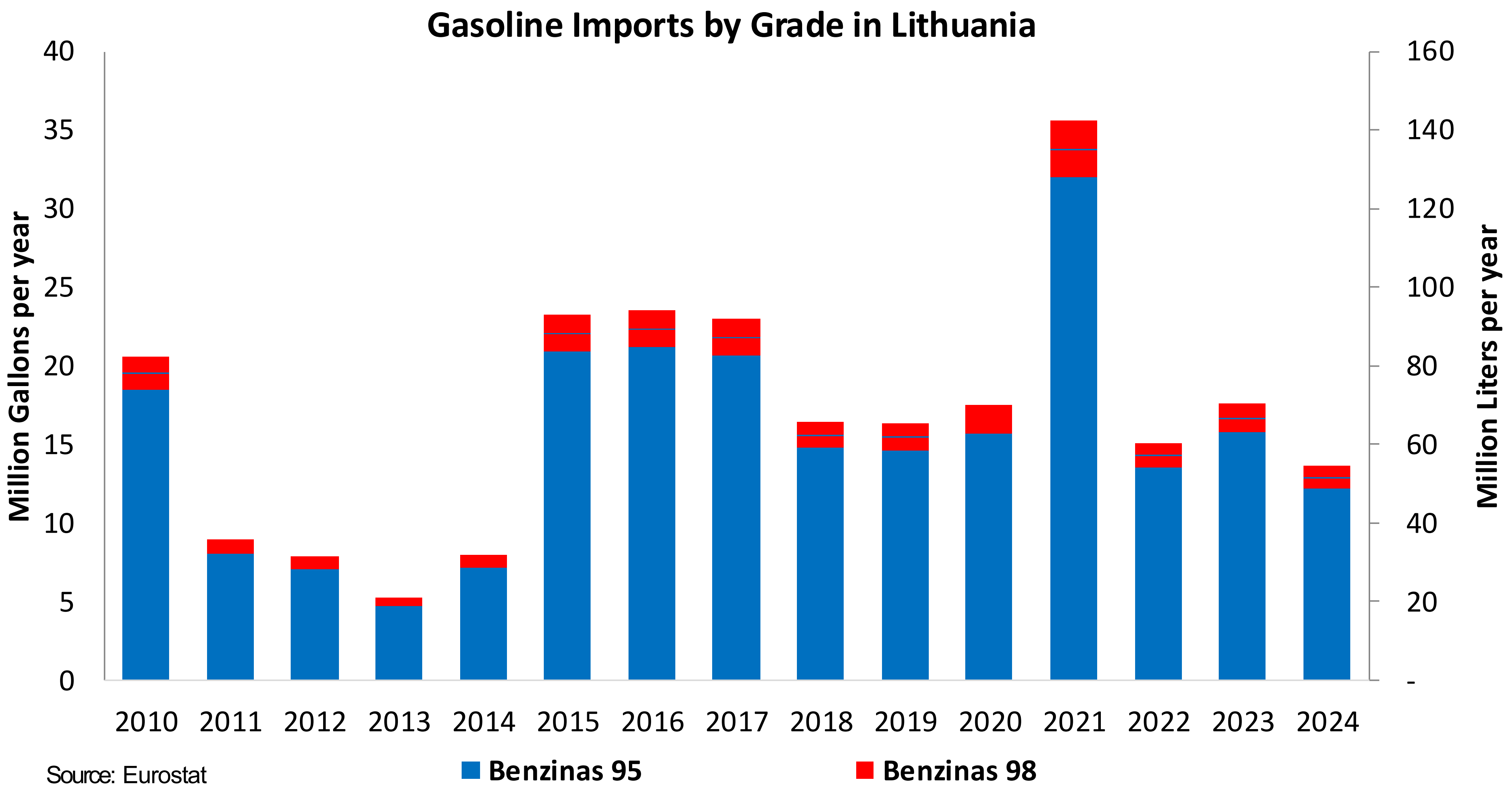
Source: Eurostat

Gasoline Production in Lithuania



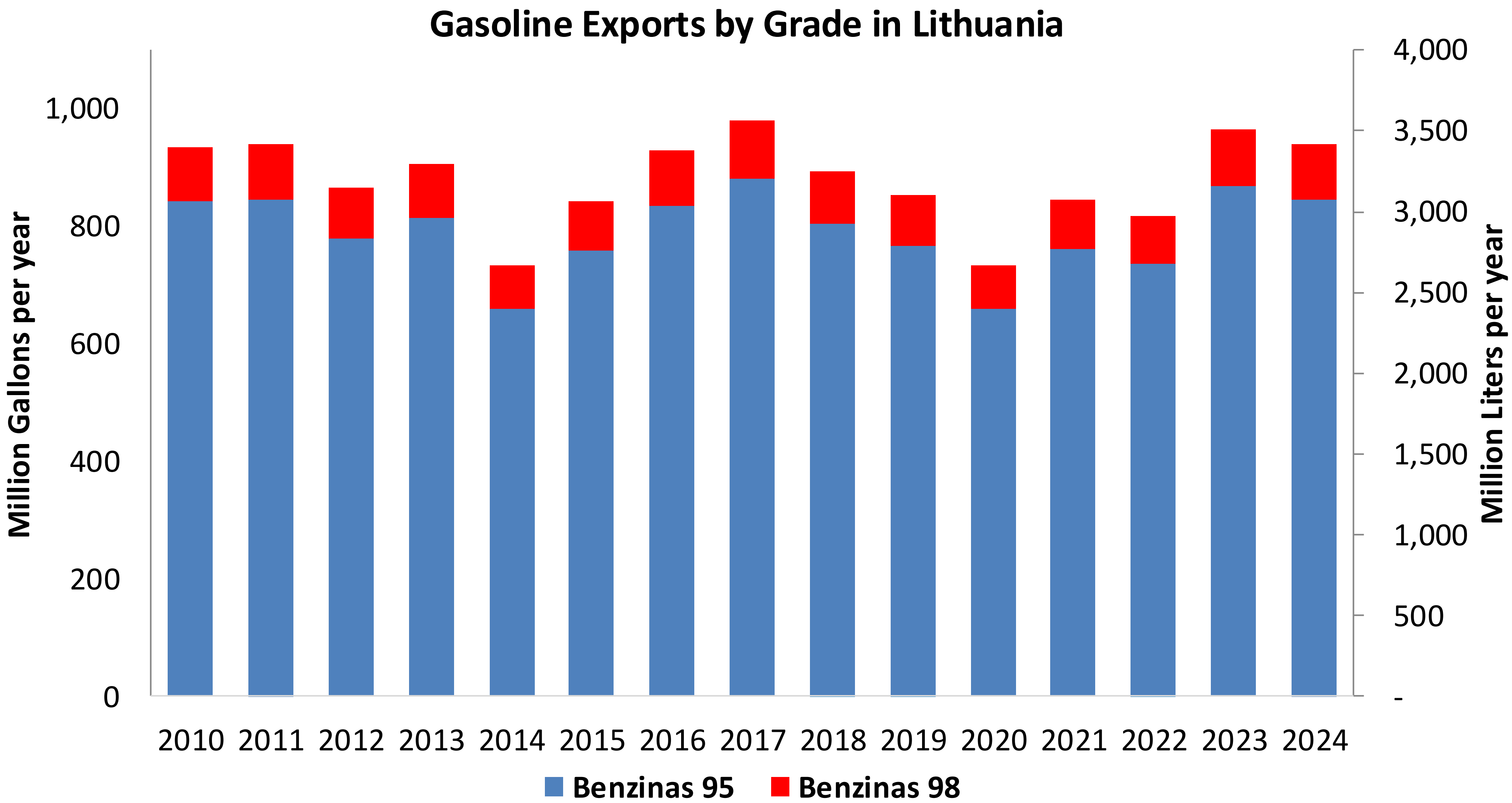
Source: Eurostat

Gasoline Imports in Lithuania



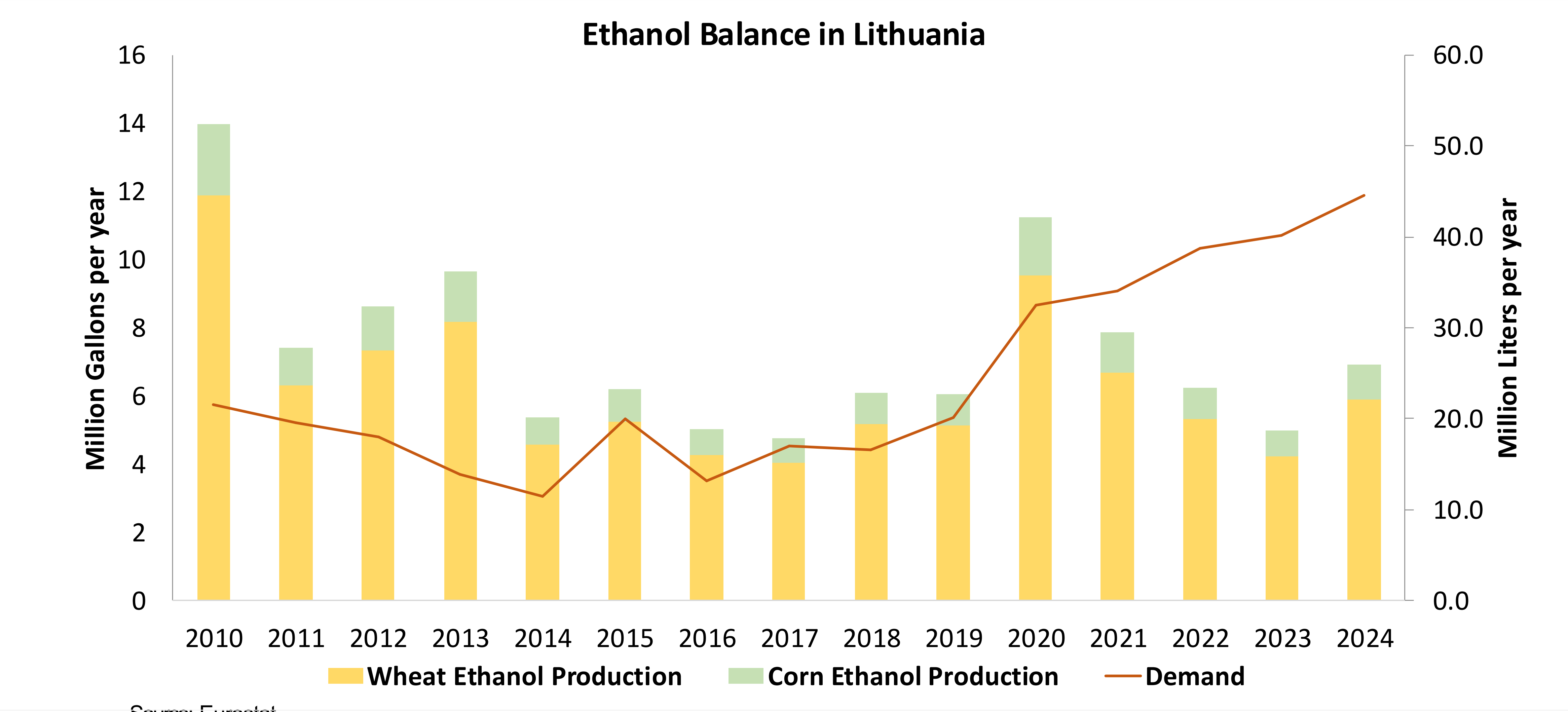
Source: Eurostat

Gasoline Exports in Lithuania



Source: Eurostat

Ethanol Balance in Lithuania



Source: Eurostat

Gasoline Quality in Lithuania

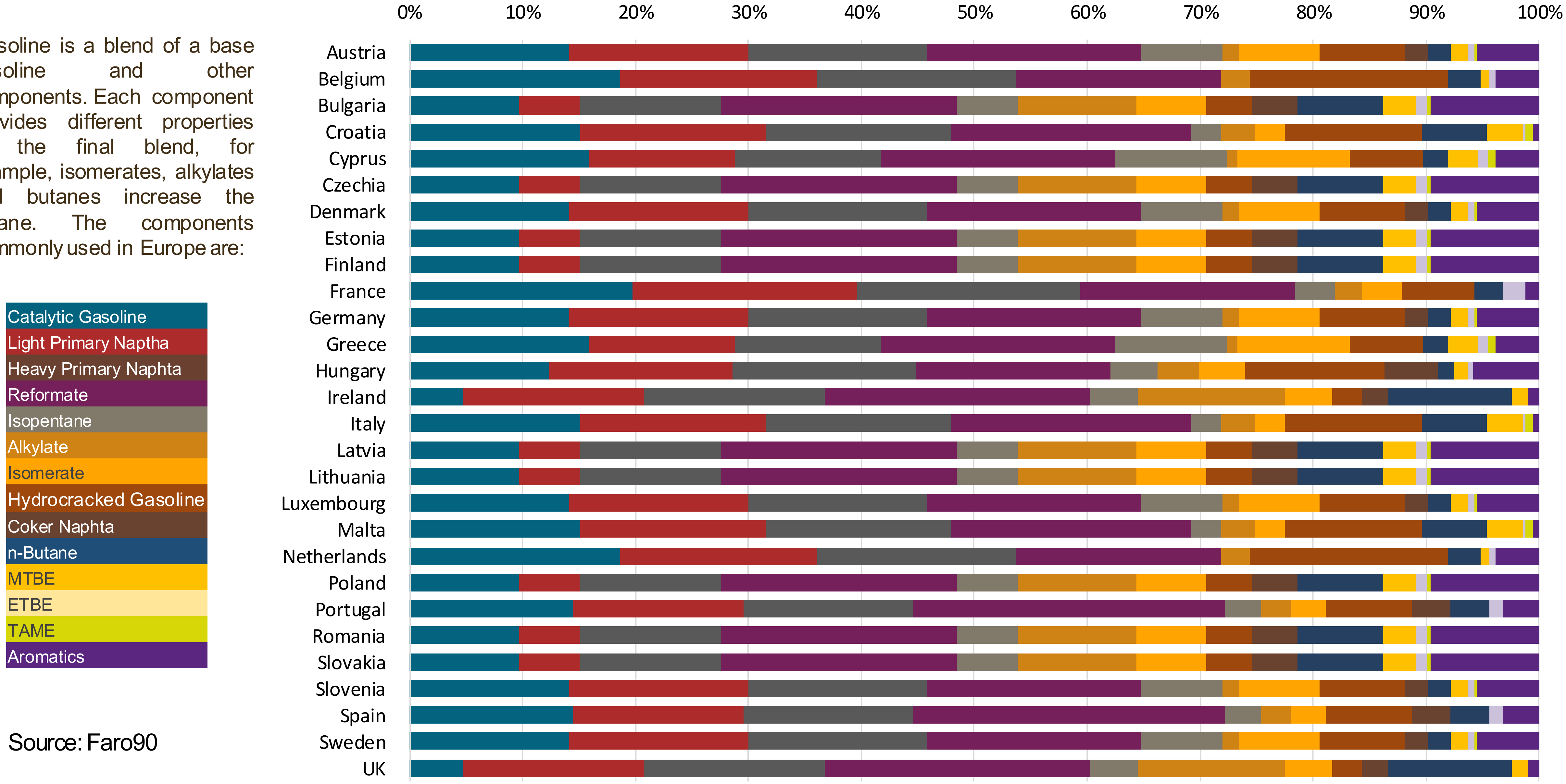
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.6							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

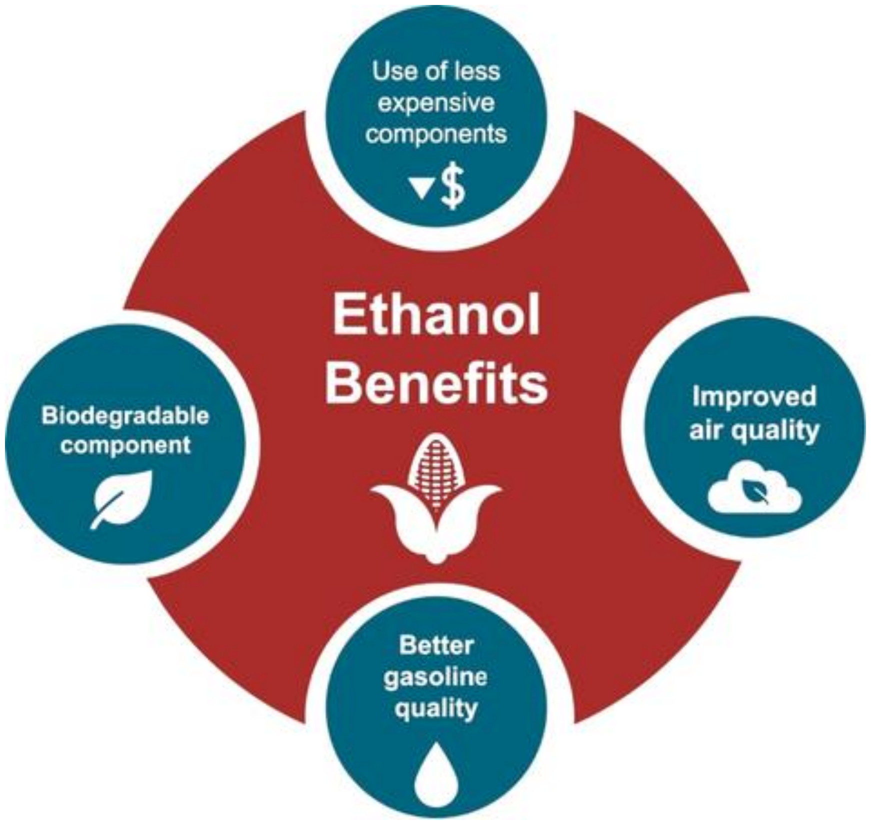
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

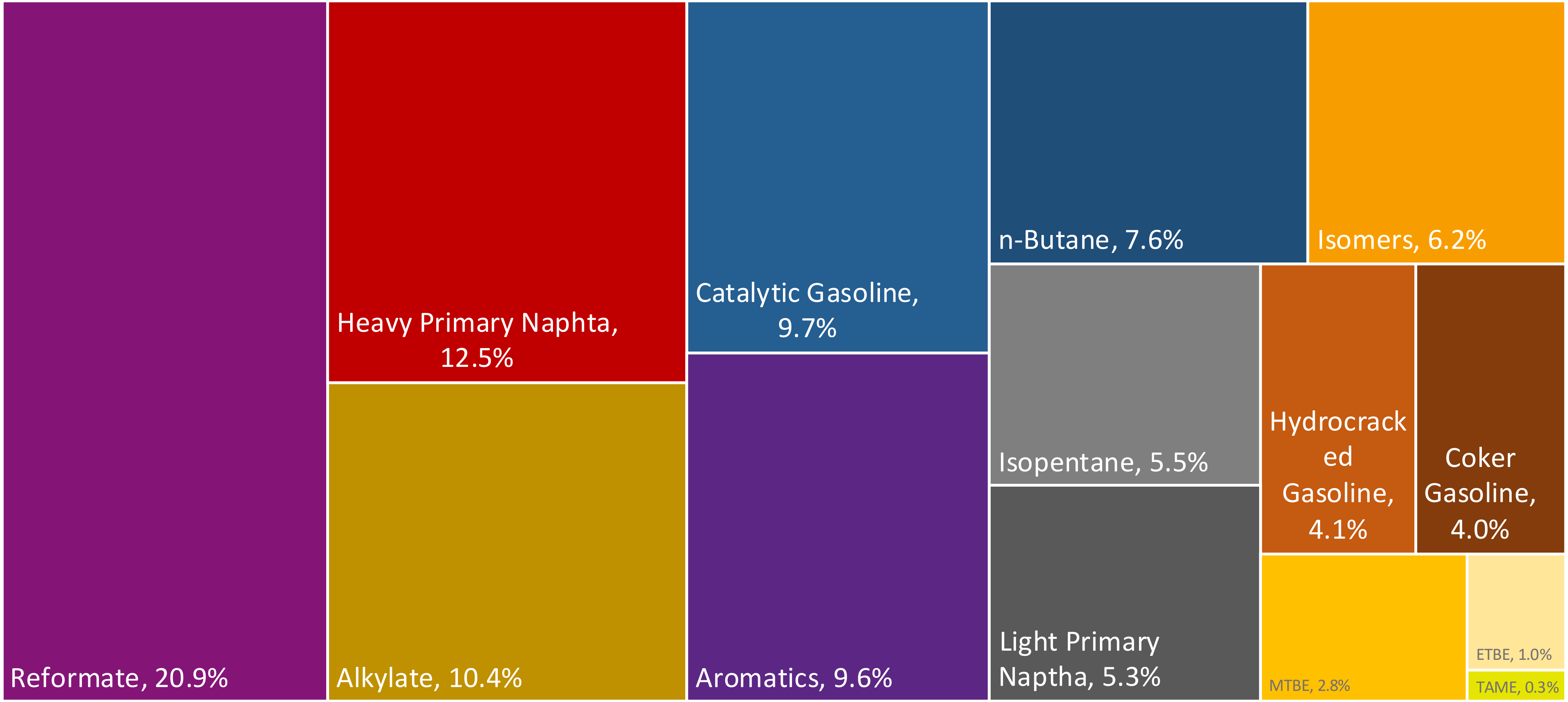
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

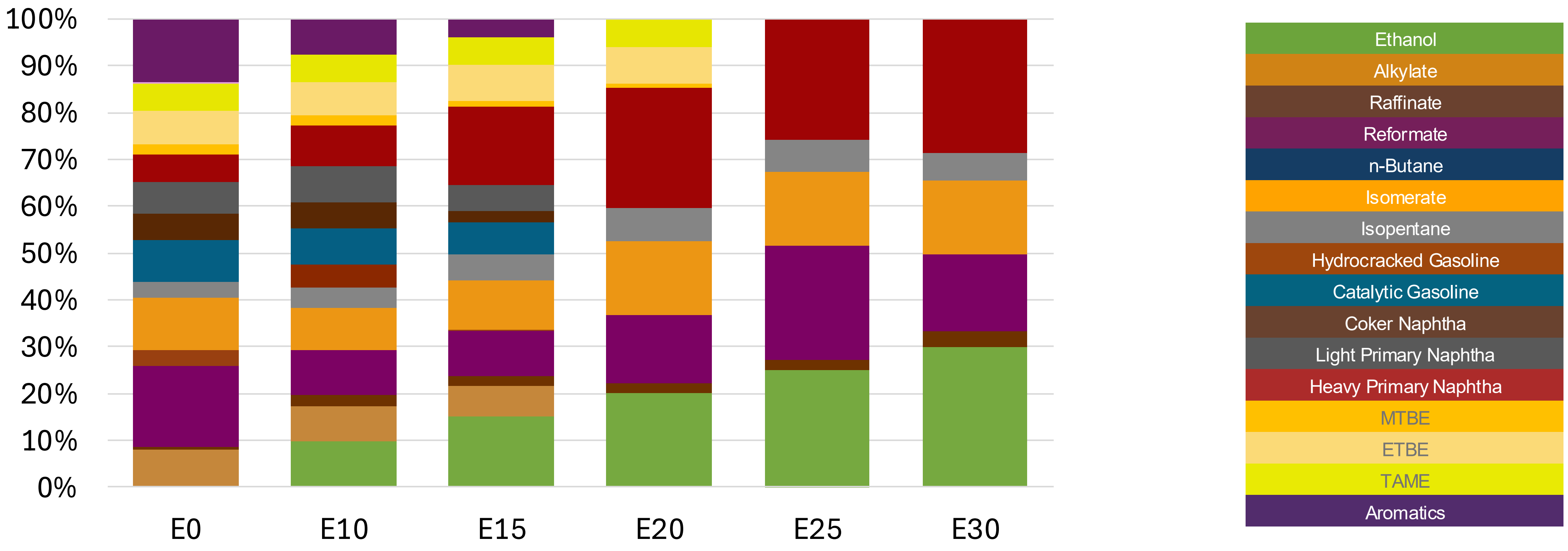


Lithuania Available Blending Components



Source: Faro90

Lithuania – Gasoline 95 – Constant Octane

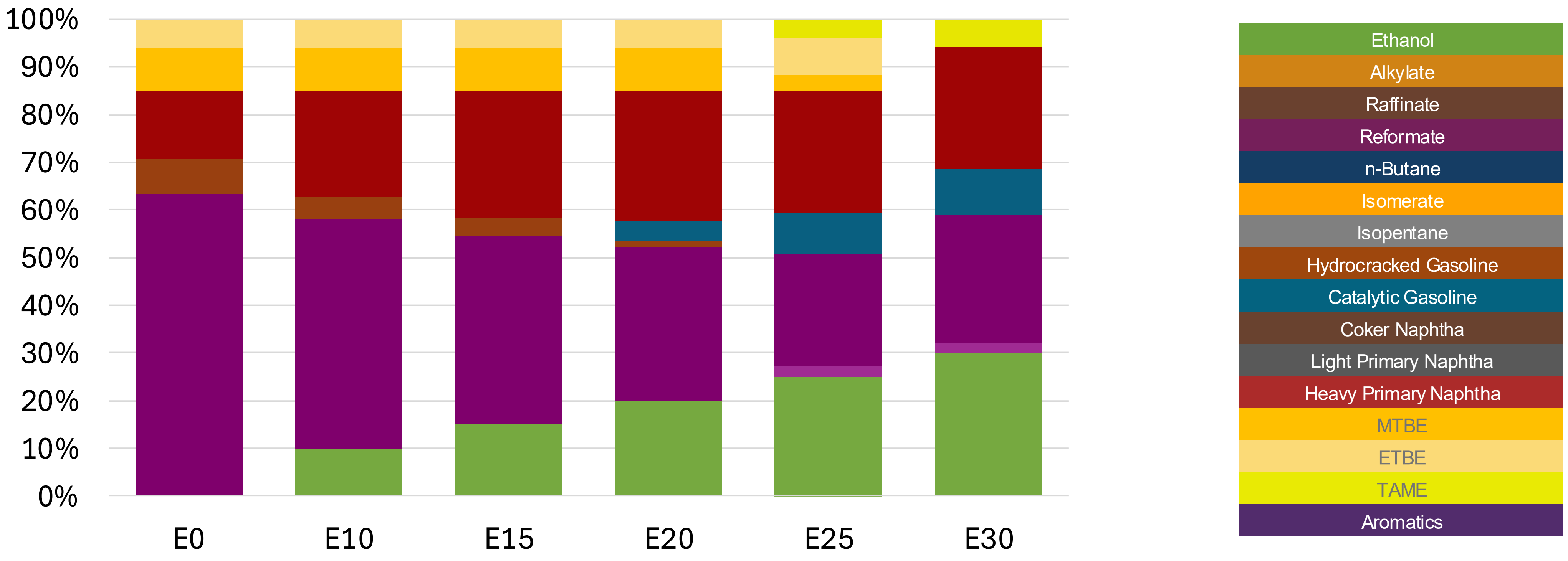


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.274	$ 2.136	$ 2.036	$ 1.921	$ 1.875	$ 1.822

Average CIF 2024 Prices

Source: Faro90

Lithuania - Gasoline 98 - Constant Octane

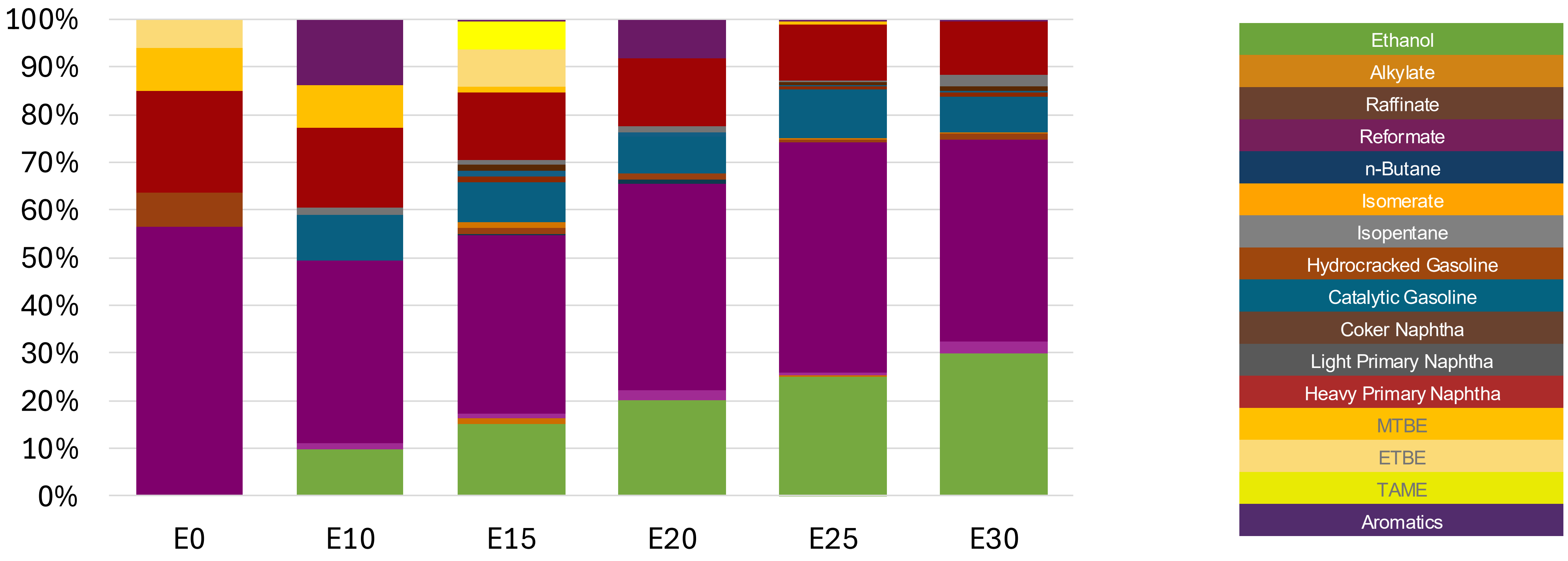


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.294	$ 2.177	$ 2.108	$ 2.036	$ 1.971	$ 1.921

Average CIF 2024 Prices

Source: Faro90

Lithuania – Gasoline 95 – Increased Octane

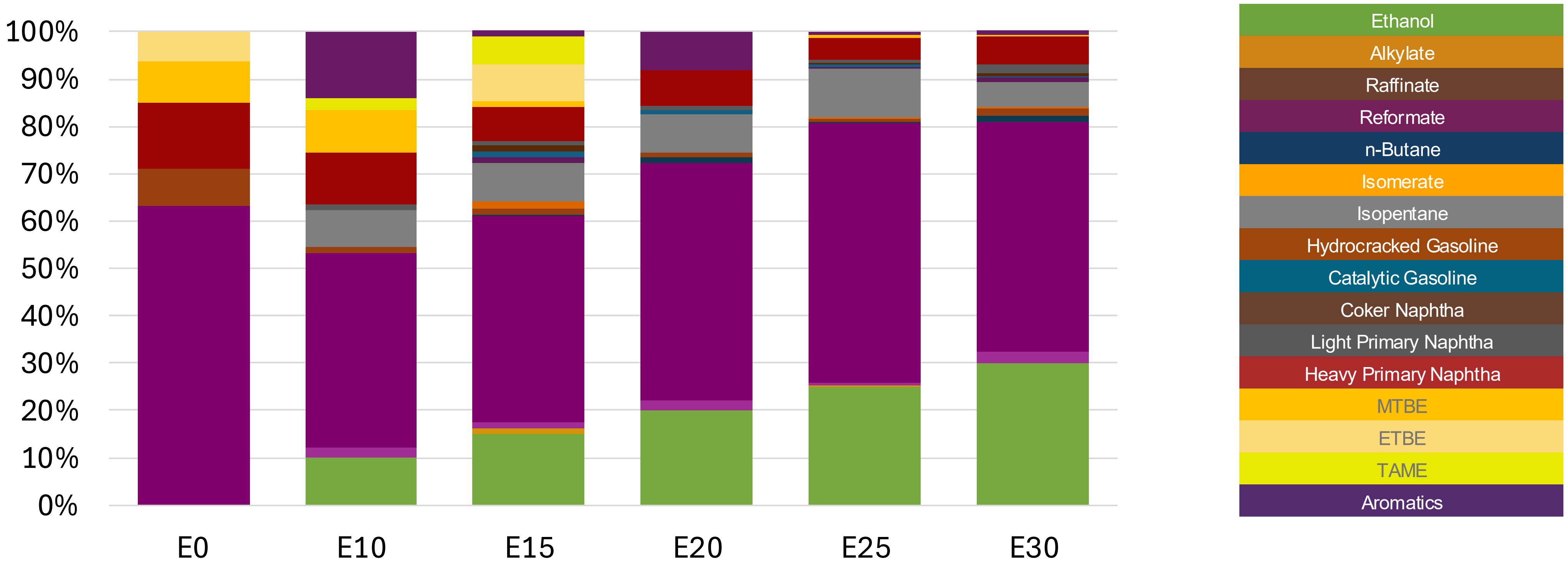


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162

Average CIF 2024 Prices

Source: Faro90

Lithuania – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

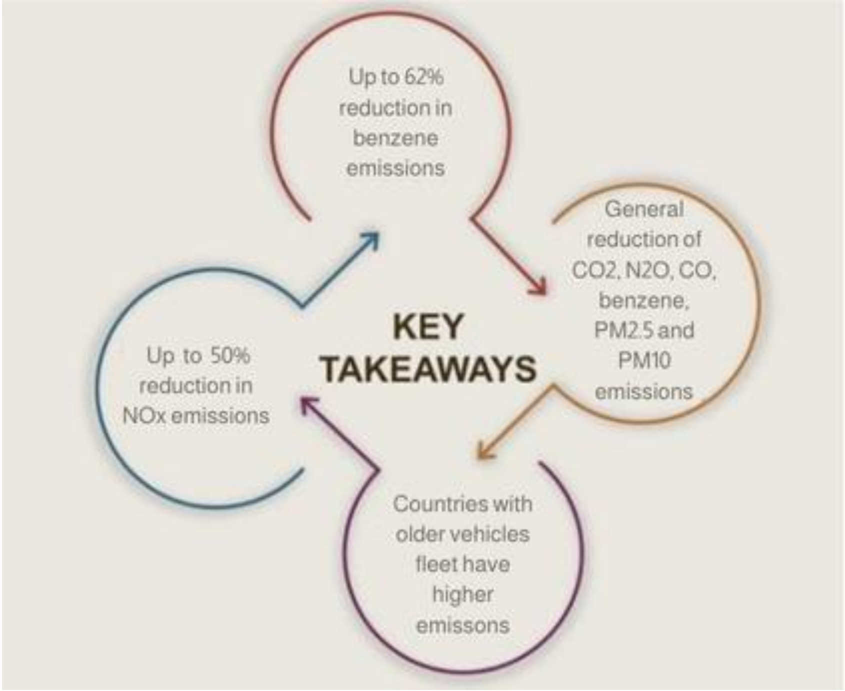
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Lithuania

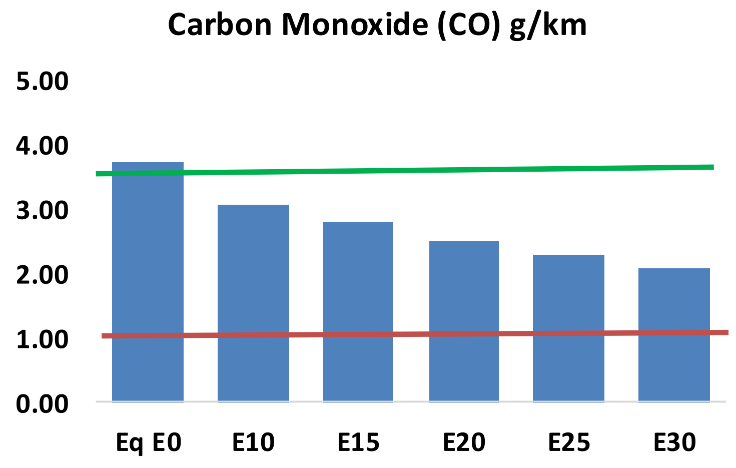
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	790
		Single FI	3-Ways		>8 years	18,498
		Multi FI	3-Ways		>8 years	277,557
			Euro III		>8 years	160,441
			Euro IV		>8 years	374,388
			Euro V		>8 years	131,925
			Euro 6		>8 years	388,917
					4-8 years	404,463
					<4 years	187,055
			Hybrid		All	166,600
		Electric			All	24,698
	Medium	Carburator	3-Ways	PCV Tank	>8 years	16
		Single FI	3-Ways		>8 years	500
		Multi FI	3-Ways		>8 years	31,495
			Euro III		>8 years	32,274
			Euro IV		>8 years	128,169
			Euro V		>8 years	40,697
			Euro 6		>8 years	129,639
					4-8 years	183,715
					<4 years	146,617
			Hybrid		All	116,948
		Electric			All	19,121
Sml Engines	Light	4-Cycle	None	PCV	>8 years	99
			Euro III		>8 years	869
			Euro IV		>8 years	11,393
			Euro V		>8 years	5,193
			Euro 6		>8 years	13,375
					4-8 years	10,222
					<4 years	6,158
			Hybrid		All	5,088
		Electric			All	786
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	15,344
		Fuel-Injection	3-Ways		>8 years	101,246
			Euro III		>8 years	34,226
			Euro IV		>8 years	106,050
			Euro V		>8 years	42,548
			Euro 6		>8 years	72,206
					4-8 years	79,325
					<4 years	78,889
			Hybrid		All	19
		Electric			All	907

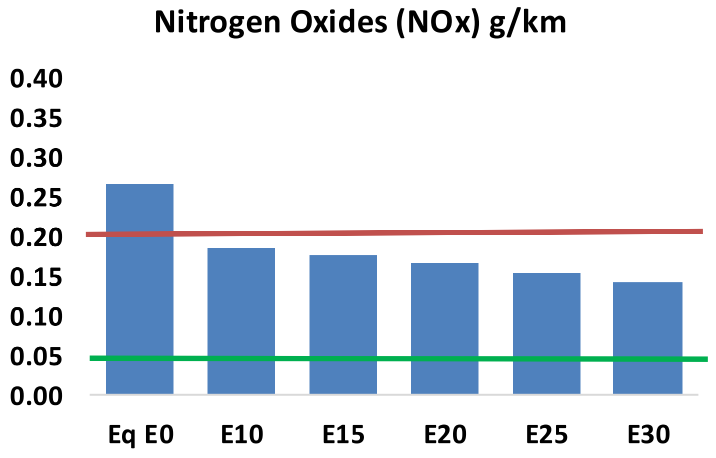
Vehicle Fleet: 3,548,467
Gasoline Vehicle Fleet 745,407

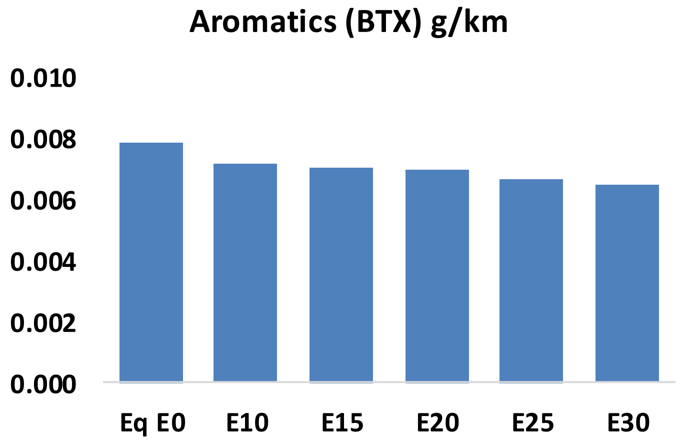
Source: Eurostat, ACEA

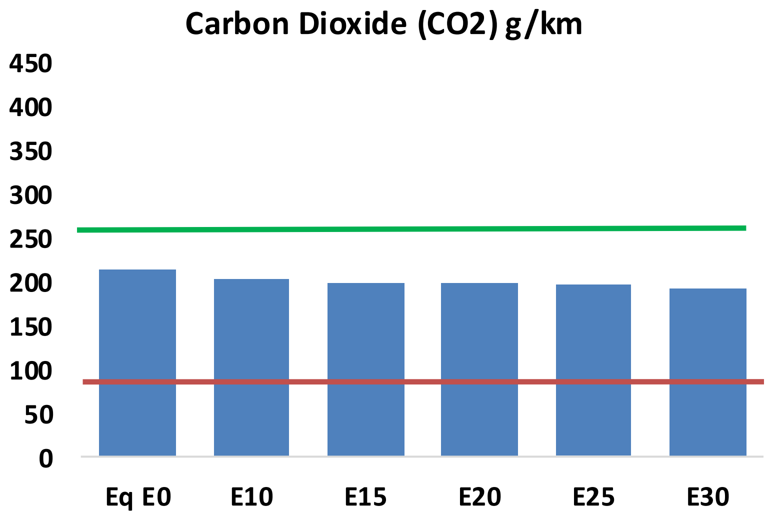
Lithuania – Vehicular Emissions

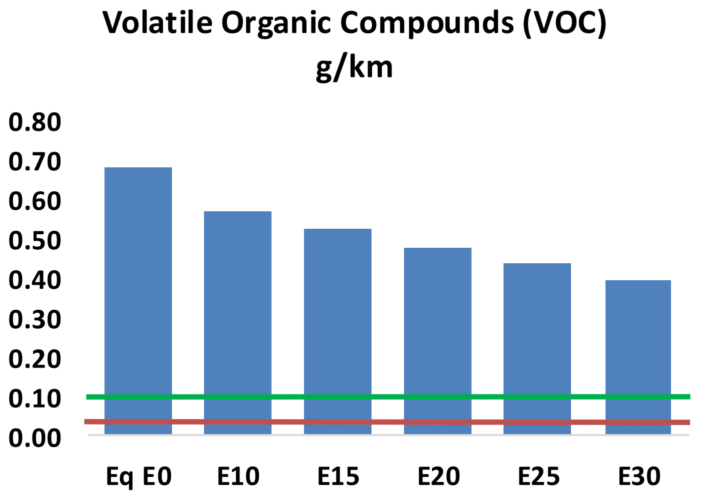
TIER III BIN 125 United States
Euro 6 Standard

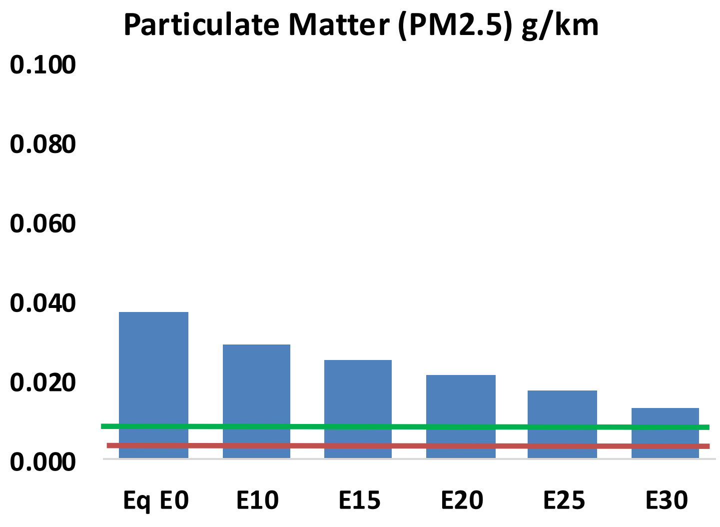












Source: Faro90

Lithuania – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	71,126	64,920	58,715	53,486	48,158	44,041	39,682	-17%	-32%
CO2	4,091,103	3,985,903	3,886,547	3,808,407	3,769,741	3,747,441	3,678,365	-5%	-8%
VOC	13,034	11,965	10,896	10,007	9,106	8,411	7,490	-16%	-30%
NOx	5,080	3,810	3,556	3,352	3,161	2,950	2,728	-30%	-38%
SOx	49	45	42	39	35	32	29	-15%	-28%
NH3	1,503	1,488	1,473	1,480	1,497	1,508	1,518	-2%	0%
N2O	69	68	68	68	70	70	71	-1%	2%
CH4	2,644	2,644	2,644	2,684	2,742	2,792	2,840	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	67	64	62	60	58	57	55	-8%	-14%
Acetaldehyde	210	242	354	540	735	840	993	68%	249%
Formaldehyde	773	752	876	1,029	1,078	1,193	1,298	13%	39%
Benzene	151	145	137	135	134	128	124	-9%	-11%
PM2.5	252	224	196	170	144	117	89	-22%	-43%
PM10	463	411	359	312	265	215	163	-22%	-43%

Source: Faro90

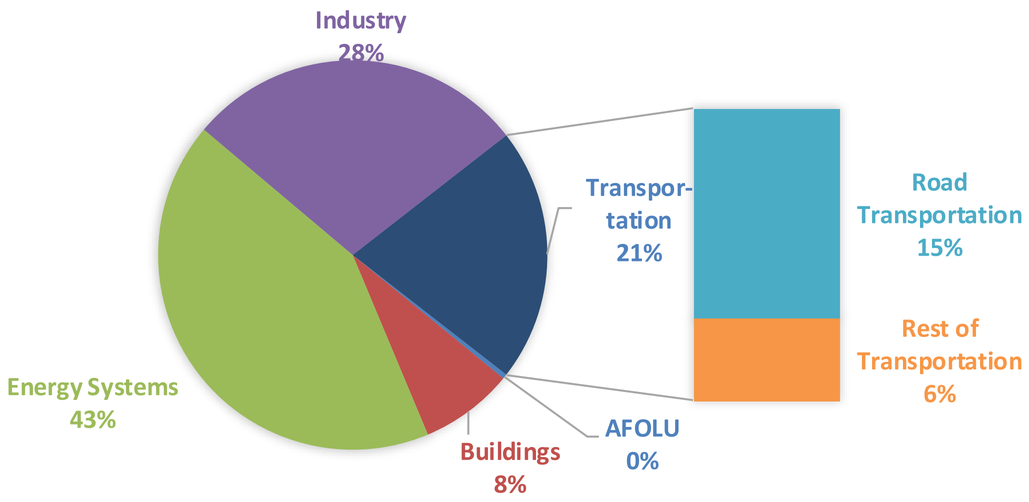
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

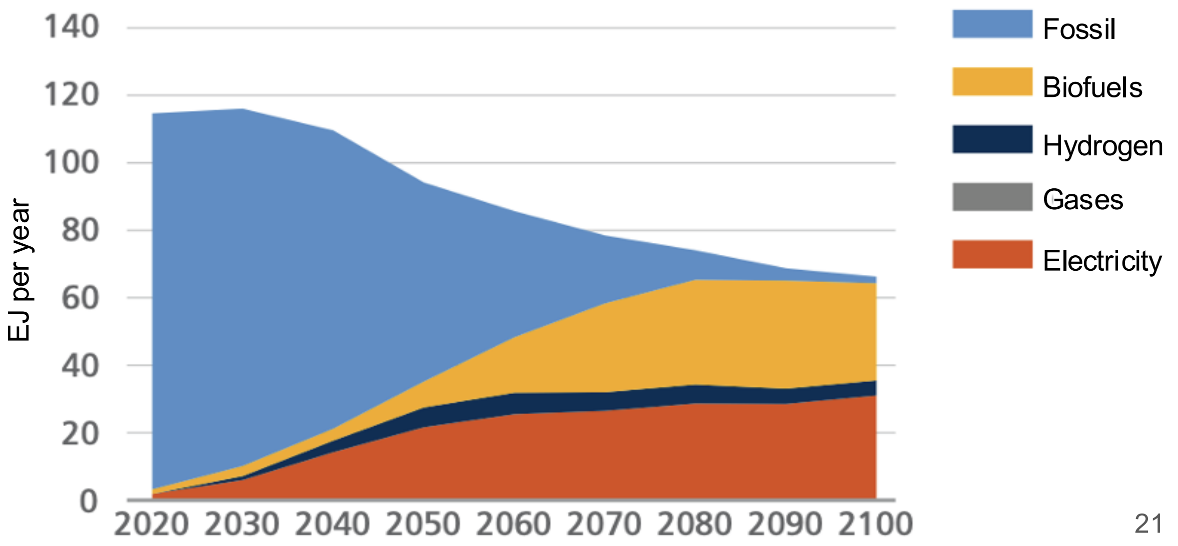
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

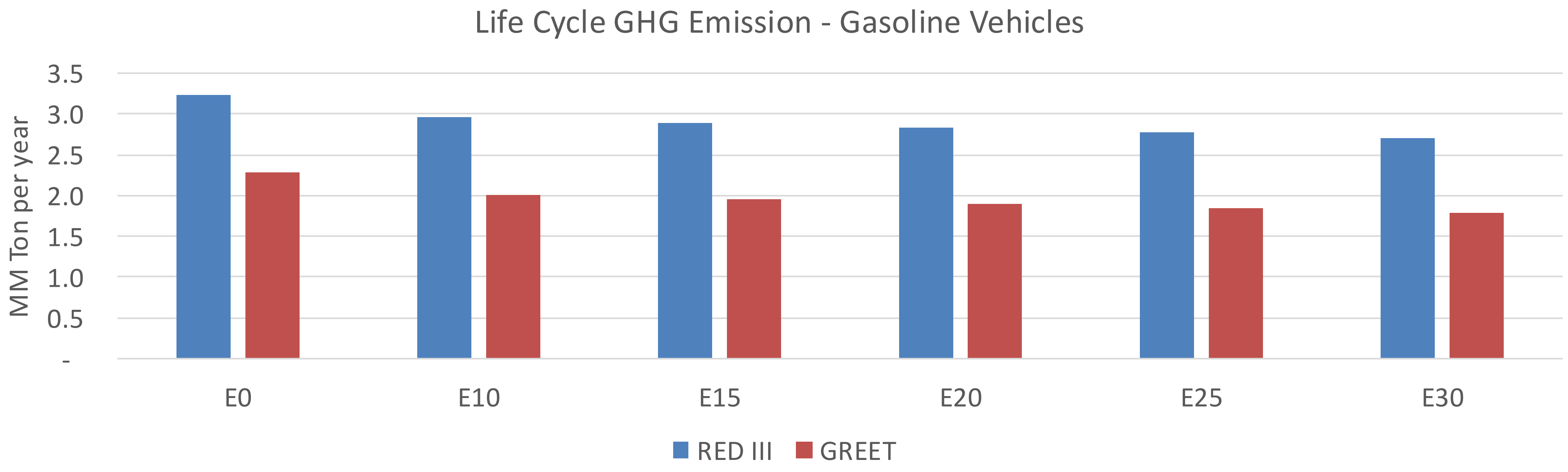
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Lithuania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	25.4
Target @ 2035 (MMT/year)	14.3
Delta	11.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.9%	14.9%	17.2%	19.3%	22.3%
RED II/III	0.0%	8.8%	11.0%	12.8%	14.4%	16.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.1%	3.5%	4.0%	4.6%
RED II/III	0.0%	2.6%	3.2%	3.7%	4.2%	4.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90